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1. Airfoil Performance

The aerodynamic performance of an airfoil is strongly influenced by its thickness, camber and leading edge geometry. For wind energy and small scale aerodynamic applications understanding how these features affect lift, drag and stall behaviour is essential. In this project the aerodynamic characteristics of three airfoil profiles (NACA 0019, NACA 2421 and a circular profile) are evaluated using different methods, such as thin airfoil theory, Xfoil, and steady state 3D computational fluid dynamics (CFD) simulations.

1.1. Airfoil Selection and Rationale

The following profiles are analysed:

NACA 0019 – the profile is a symmetric airfoil with thickness equal to 19% of the chord length and zero camber, this is selected because the symmetric profile can provide a clean baseline for assessing how adding camber affects aerodynamic performance.

NACA 2421 – this profile has a max camber of 2% of the chord and the location of the max camber is at 0.4c, with 21% thickness. This is chosen to represent a highly cambered profile to investigate effects of camber on lift, drag and stall,

Circular profile – The circle is chosen as a bluff body, which has no conventional lift generation mechanism, large separation at low angles of attack (AoA) and can serve as an extreme baseline for lift, drag and stall comparisons.

1.2. Computational Setup

1.2.1. Reynolds Calculations

The computational domain is chosen based on estimated Reynolds number because at moderate values (Re = 1e4 – 1e5) laminar separation bubbles and early separation occur, whereas higher Re has a more energetic turbulent boundary layer which affects how close outer boundaries need to be to negate blockage effects or artificial pressure gradients [1].

The Reynolds number is defined as [2] :

$$Re = \frac{\varrho V d}{\mu}$$

Where ϱ is density, μ viscosity, V free stream velocity and d a characteristic length dimension. Given $V = 8m/s$, $d = 0.1m$, $\varrho = 1.2kg/m^3$ and $\mu = 1.8 * 10^{-5} kgm^{-1}s^{-1}$:

$$Re = \frac{1.2 * 8 * 0.1}{1.8 * 10^{-5}} = 5.33 * 10^4$$

This value sits firmly in the moderate Re region where the airfoil acts differently on its pressure and suction sides. The boundary layer on the pressure side typically remains laminar attached,

however, the suction side has an adverse pressure gradient that can cause laminar separation. If the separated layer transitions to turbulence sufficiently upwards of the trailing edge, the turbulent flow can reenergise the boundary layer allowing it to reattach, this forms a laminar separation bubble (LSB), between the separation and reattachment points [1], [3]. LSBs can decrease performance and increase sensitivity to small changes in AoA, however can trigger early transition to turbulence stabilising the boundary layer and preventing large scale separation [4].

1.2.2. Domain and Boundary Conditions

Jardin et al [4] did a CFD study on a NACA 0012 airfoil at Re of 53,000 and used a domain of length 50c (10c upstream, 40c downstream), height of 20c and width of 0.2c to capture 3D effects sufficiently at low AoA, and showed good agreement with their experimental data.

Because this project will use steady state rather than unsteady simulations, less downstream length is required as time dependant wake structures aren't tracked leaving artificial reflections or back-pressure the main reasons for an extended downstream. This project uses a domain of 10c upstream, 20c downstream, 20c height, span of 0.2c, and uses a C-grid with the leading edge tip located in the centre and the downstream part is square. The domain is shown in figure 1.

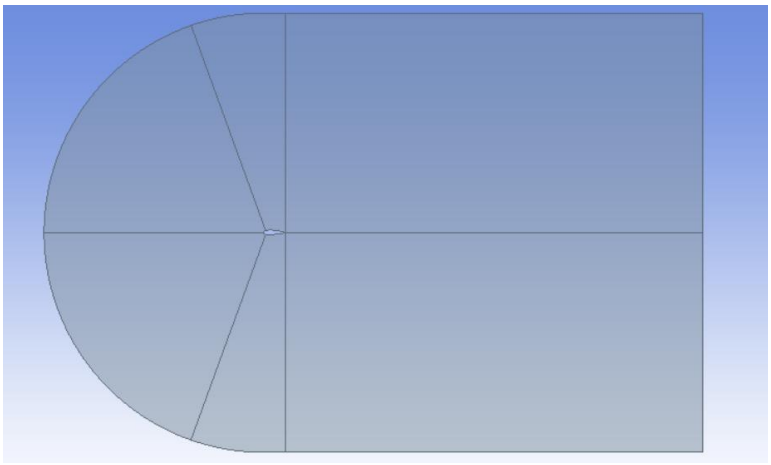


Figure 1 - Computational Domain

At the inlet a uniform velocity of 8m/s, turbulence intensity of 0.15% and turbulence length of 0.1c are used to represent low turbulence inflow, levels upwards of 2% can strongly effect LSB formation [4]. The outlet set to 0 gauge pressure and the top and bottom boundaries are zero shear to prevent unwanted boundary effects. The airfoil wall has a non-slip wall condition with zero roughness as this can strongly affect transition [5].

1.3. Analytical Lift and Drag Estimates

Xfoil is used to obtain each airfoil's lift (Cl) and drag (Cd) coefficient, a purely analytical method, thin airfoil theory, is used for comparison. Thin airfoil gives:

$$C_l = 2\pi(\alpha - \alpha_{L=0})$$

Where α is the angle of attack (AoA) and $\alpha_{L=0}$ is the zero lift AoA meaning $\alpha_{L=0} = 0$ for symmetrical airfoils [6]. For the NACA 2421 $\alpha_{L=0}$ is taken from the Xfoil results as 0.7°. Thin airfoil theory assumes inviscid flow meaning Cd=0. Figure 2 shows Cl vs AoA:

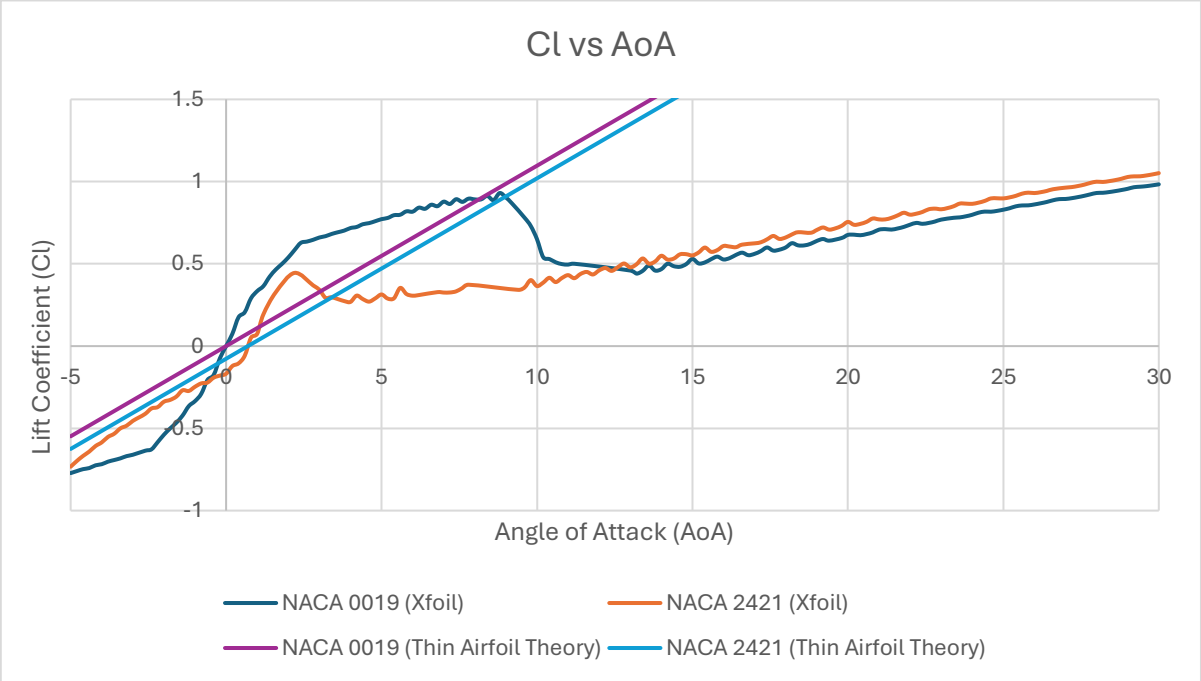


Figure 2 – Lift Coefficient vs Angle of Attack for NACA 0019 and NACA 2421

Figure 2 shows the variation between the Xfoil and thin airfoil theory, it can be observed thin airfoil theory shows a linear increase in Cl with no stall, whereas, Xfoil captures stall and nonlinear effects due to viscous effects. The discrepancy increases further at high AoA since thin airfoil theory doesn't account for separation, stall or thickness effects. Therefore, Xfoil better represents practical performance.

The symmetrical NACA 0019 shows no lift at 0° AoA, agreeing with theory, and stalls at ~8.8°. The cambered profile has zero lift at ~0.7 ° AoA showing negative lift below this. Although camber normal generates a positive lift at 0° a pressure side laminar separation bubble can reduce effective camber, which could explain this behaviour [7], [8]. At Re $1x10^5$ Xfoil predicts positive Cl at 0°. The cambered airfoil also stalls earlier and shows lower Cl, which could be due to low-Re laminar separation under the cambered profile's adverse pressure gradient [9].

Figure 3 shows the drag coefficient for the NACA0019 is low and nearly constant which is typical for airfoils operating in attached flow before stall [10]. The cambered airfoil's rise is more gradual which again suggests early laminar separation. At high AoA both profiles converge as separated flow dominates.

Figure 4 shows lift to drag ratio with NACA0019 a peak lift to drag ratio of ~23 at an AoA of 6.2°, and the cambered reaching ~8.2 at 2.2°. Each show a rapid rise followed by a sudden drop in aligning with their stall point. From this it can be observed in this Reynolds region the NACA 0019 has higher aerodynamic efficiency however has a faster drop after stall possibly giving NACA 2421 smoother performance.

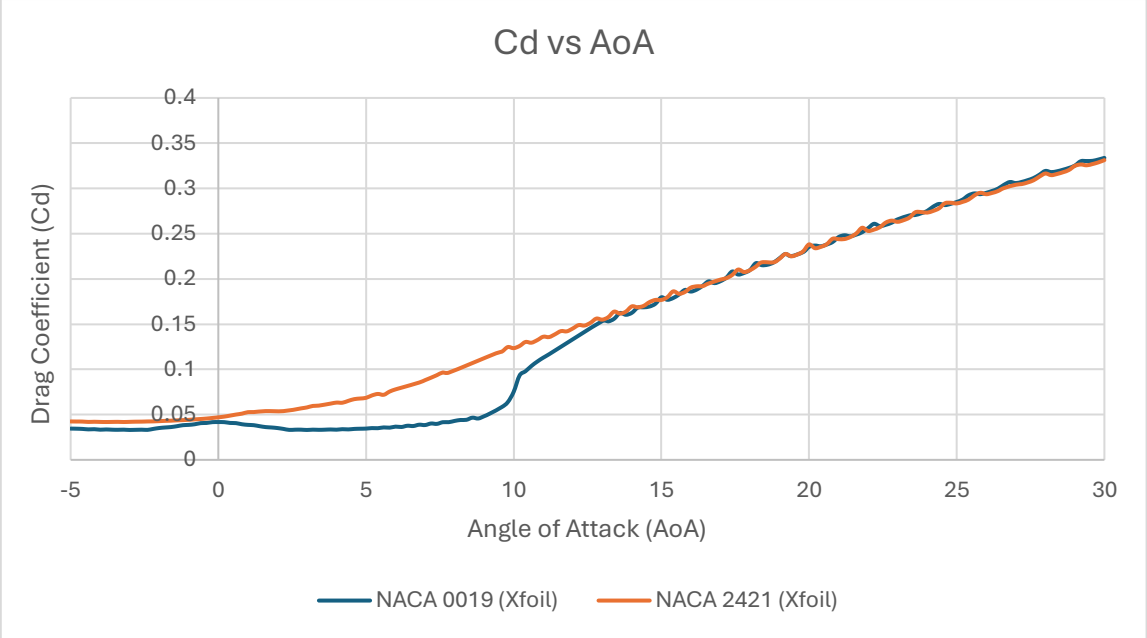


Figure 3 – Drag Coefficient vs Angle of Attack for NACA 0019 and NACA 2421

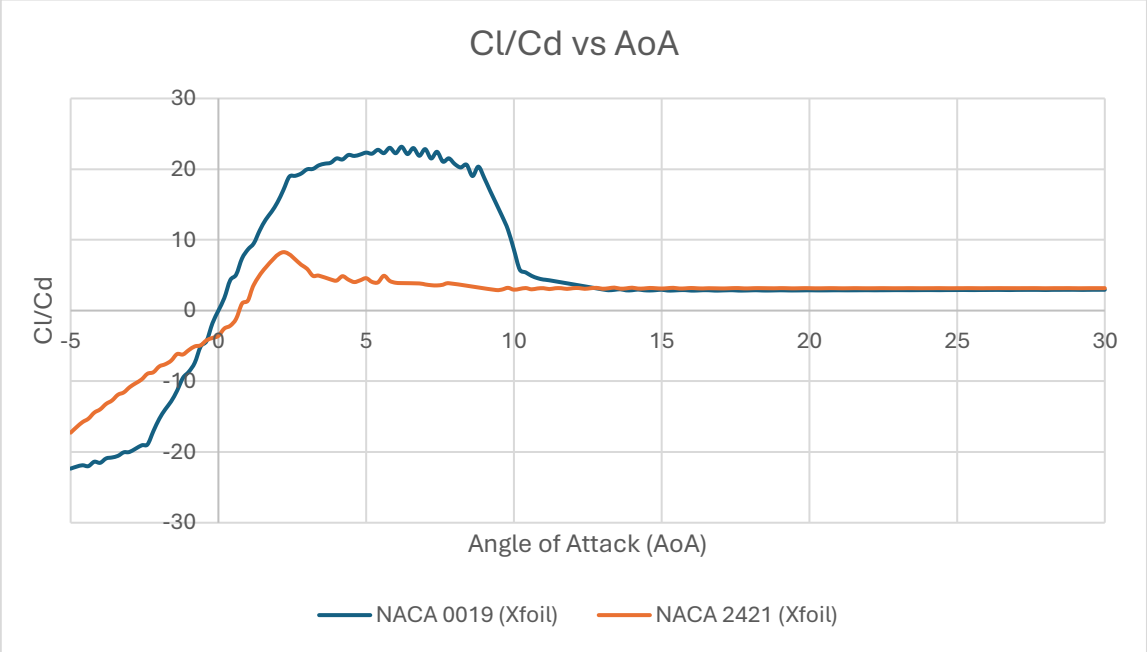


Figure 4 - Lift to Drag Ratio vs Angle of Attack for NACA 0019 and NACA 2421

1.4. CFD Simulations

Simulations for each profile are performed in steady state to reach a time averaged solution at a lower computational cost, however, low Reynolds numbers with time dependant phenomena like vortex shedding or separation bubbles may not reach convergence without time accurate RANS solvers [11]. The $k-\omega$ SST model with the gamma algebraic ($\gamma - Re_\theta$) transition model is used as accurately predicts where aerodynamic stall occurs [3]. Figure 5 shows the structured mesh which is created with face meshing and biases to maintain a y+ value of < 1

around the airfoil, which is essential for the flow to be accurately resolved near to the wall [4]. The mesh has an average and minimum orthogonal quality of 0.97 and 0.25 and average and maximum skewness of 0.05 and 0.15 respectively, these values sit comfortably within Ansys best practices [12].

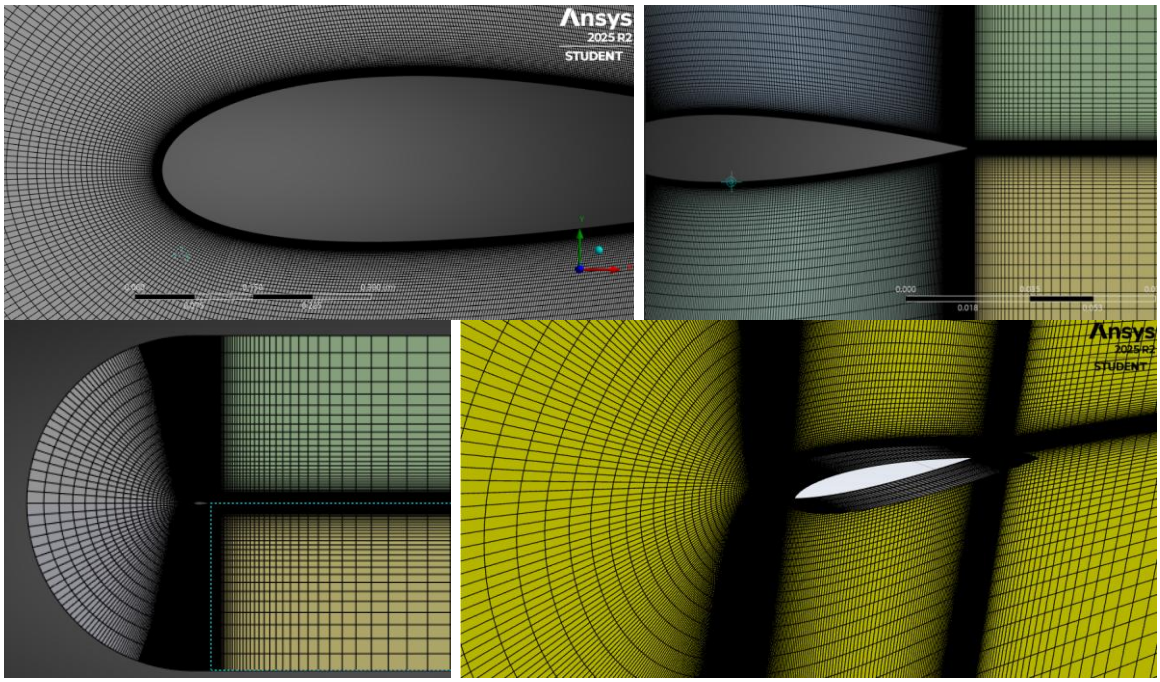


Figure 5 – Mesh around NACA 0019

Convergence was reached when residuals dropped below $1x10^{-6}$ which typically took around 1000 iterations with 300,000 elements. Figure 6 shows a pressure contour of the airfoil at 9° AoA, with high pressure (red) forms where the flow stagnates at the lower leading edge, low pressure over the upper surface due to rapid acceleration of flow which causes a pressure drop, and there is a return to normal pressure at the trailing edge. The pressure difference between the upper and lower surfaces is where the lift is generated, showing the general physics are being captured by the simulation.

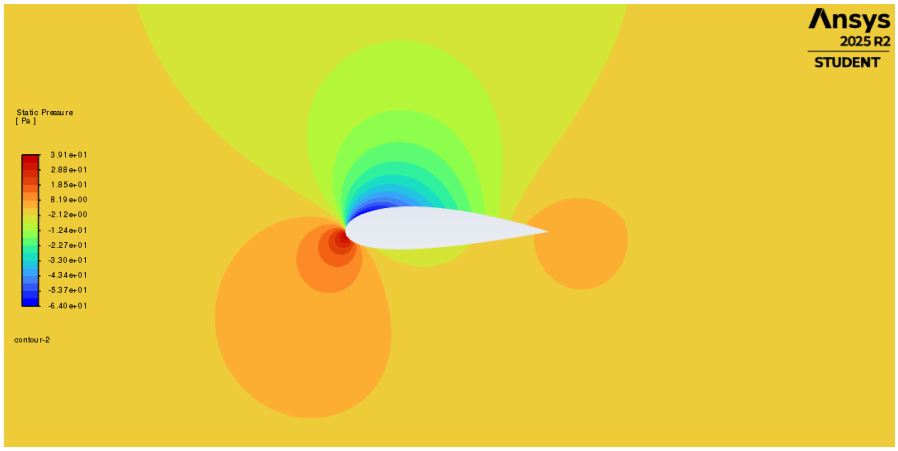
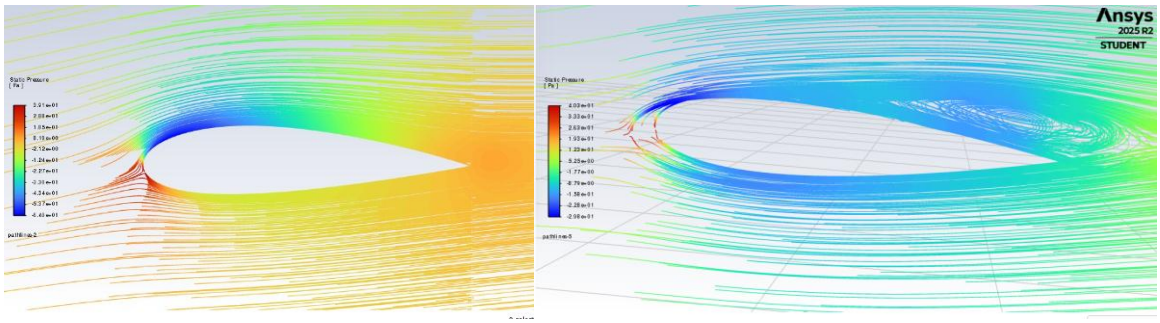


Figure 6 - Pressure Contour for NACA 2421 at 9deg AoA

Figure 7a shows the and 7b show the path lines of the airfoil pre and post stall. Pre stall the flow remains attached to the airfoil surface with the path lines smoothly following the contour, in contrast the post stall shows substantial flow separation with path lines looping and swirling in vortices instead of following the surface.



(a) (b)

Figure 7 - Path lines Over Airfoil Pre and Post Stall

The following figures show the lift and drag coefficients from CFD compared with Xfoil for each airfoil profile:

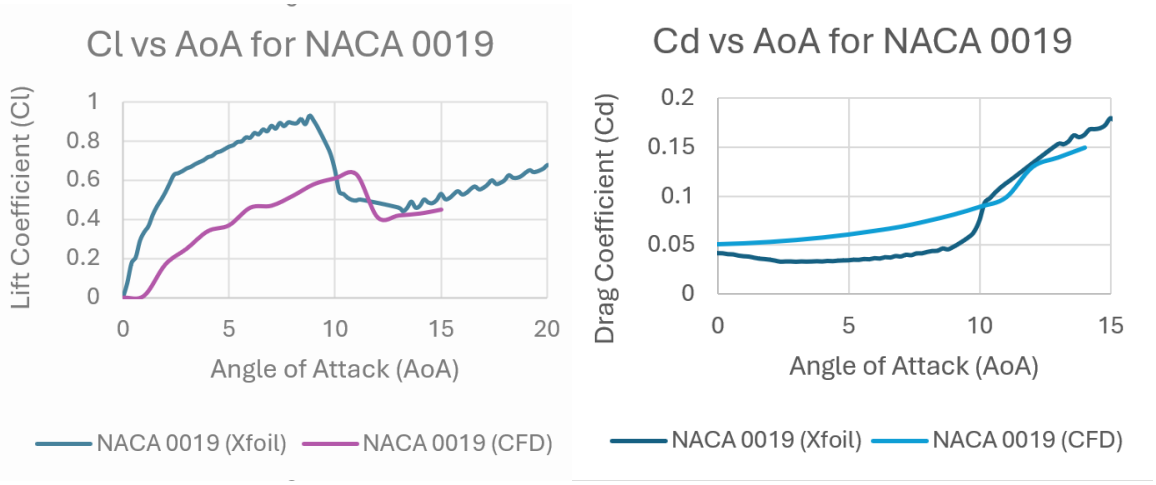


Figure 8 - NACA 0019 Cl and Cd vs AoA

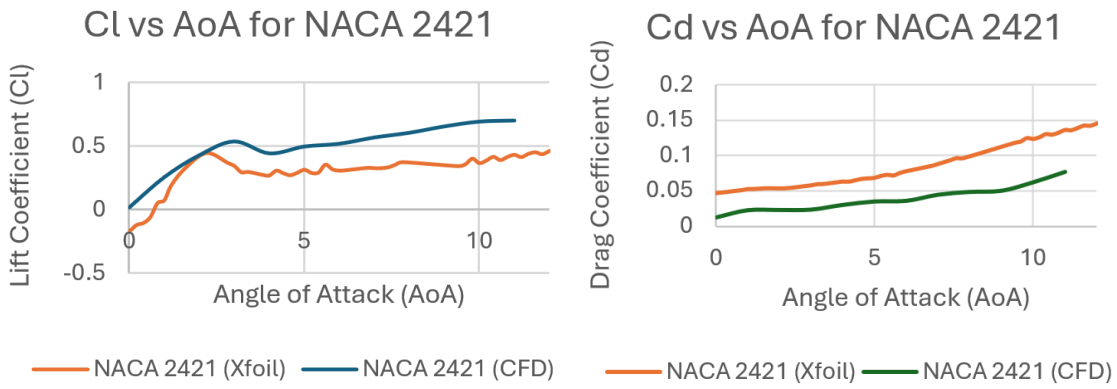


Figure 9 - NACA 2421 Cl and Cd vs AoA

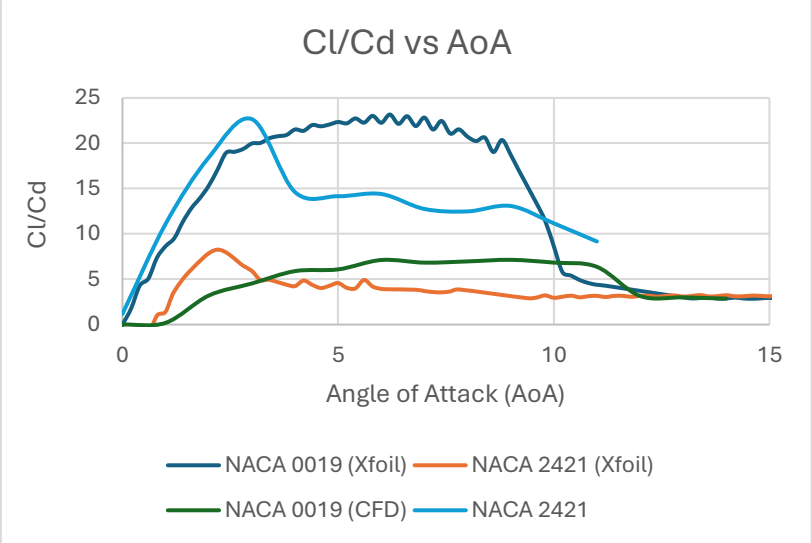


Figure 10 - Lift to Drag Ratio for NACA 0019 and NACA 2421

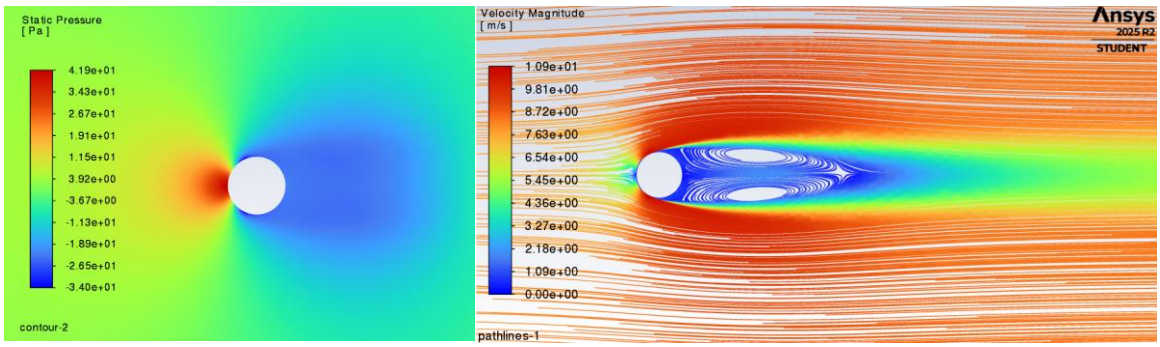


Figure 11 - Pressure Contour and Path lines for Circular Profile

AoA	Cl	Cd
0	-0.0049	0.787
5	-0.0014	0.791
10	0.0031	0.789
15	0.0082	0.790

Table 1 - Variation of Cl and Cd with AoA for Circular Profile

1.5. Final Comparisons

The NACA 0019 CFD shows a decrease in peak lift and a slight increase in drag when compared to Xfoil results. As discussed in the Reynolds number section, low Reynolds flow is prone to LSB's and early flow separation, both Xfoil and the CFD, setup with the $\gamma - Re_\theta$ transition model, do simulate the boundary layer with the laminar to turbulent transition, however the CFD transition model should predict small LSBs that Xfoil may not depending on empirical correlations [13]. Stall is captured in both, confirming CFD is following the trend of an airfoil varying with AoA, however, the stall point is slightly shifted forward by ~2°. Experimental data for each profile should be checked to determine the most accurate method, however data on

the NACA 0019 or 2421 at low Reynolds numbers is very scarce. NACA 2421 show similar behaviour with stall predicted slightly above (~1°), however, this time has higher lift and lower drag than Xfoil.

Figure 11 shows the circular profile's pressure and pathline plots, there is a high pressure region at the front where flow hits it head on, pressure drops along the sides as the flow accelerates, the flow then separates from the surface and there is a turbulent and recirculating wake creating a region of low pressure. Table 1 shows the drag is far higher than the two airfoils and is insensitive to change in AoA due to its symmetrical shape, as expected its lift is also very low however, shows a very slight increase possibly due to asymmetric flow creating small pressure differences.

CFD may provide better insight into the underlying physics through visual plots and contours, however it is very dependent on the mesh, solver settings and computational power. Xfoil has remained a widely used and trusted method of estimating lift and drag coefficients for years, computational tools such as BEM commonly use Xfoil to simulate rotating blades. Although transient, highly refined CFD may get more accurate results, the computational efficiency of tools like BEM and Xfoil is often preferred in the earlier stages of aerodynamic design.

2. Wave Theories

2.1. Airy Wave Theory

Airy wave theory rests on a number of simplifying assumptions which linearise the governing water wave problem making it useful for many engineering and physical applications. The six key hypotheses and their physical meanings are described below, the discussions are largely based on information from [14], [15]:

1. Constant water depth: The seabed (bottom) is assumed to have a limited but constant depth and a horizontal bottom, this essentialy means the seabed is fixed, impermeable and flat. The waves don't interact with a sloping or moving bottom because the model assumes these variations don't exist.

Mathematically the kinematic bottom boundary condition can be expressed as:

$$\frac{\partial \Phi}{\partial z} = 0, at\ z = -d$$

Where $\Phi(x, z, t)$ is the velocity potential, z is the vertical coordinate measured positive upwards with origin at free surface and d is the constant water depth.

2. Constant Density: The water is assumed to be incompressible and have constant density, this means that density changes from temperature or salinity occur over scales much larger than a few wavelengths, so they can be ignored locally.

Mass conversion for a fluid of constant density leads to the continuity equation:

$$\frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y} + \frac{\partial u_z}{\partial z} = 0$$

Where u_x, u_y, u_z are the velocity components in the x, y, z directions and x, y, z are spatial coordinates.

3. Negligible Viscosity: The water is assumed to have no viscosity as it behaves as an ideal fluid, these viscosity induced internal forces are typically not considered for the large wavelengths. This essentially means that no viscous dissipation affect wave motion within the water column.

This means the viscous terms can be removed from the momentum equations, the linearised momentum equation therefore results in:

$$\frac{\partial u_x}{\partial t} = -\frac{1}{\varrho}\frac{\partial p}{\partial x}$$

Where t is time, ϱ is constant water density and p is the fluid pressure.

4. Surface Tension Neglected: The only external force is gravity, surface tension isn't considered, this is excluded because the theory applies to waves longer than a few centimetres.

Mathematically this means the dynamic boundary condition sets the pressure at the free surface equal to atmospheric pressure without any surface tension term $p = 0, at\ z = 0$.

5. Linearisation: The main requirement of Airy wave theory is that the wave amplitudes are small compared with the wavelength and depth, because of this nonlinear terms can be removed to linearise the theory. This essentially means that because amplitudes are small, the actual free surface can be replaced by the mean free surface in boundary conditions.

Because the waves are small, all free surface conditions are applied at the mean surface $z = 0$ rather than the real displaced surface:

Kinematic free surface condition:

$$\frac{\partial \eta}{\partial t} = \frac{\partial \Phi}{\partial z}, \qquad at\ z = 0$$

Dynamic free surface condition:

$$\frac{\partial \Phi}{\partial t} + g\eta = 0, \qquad at\ z = 0$$

Where $\eta(x, t)$ is the free surface elevation and g is gravitational acceleration.

To satisfy the linearisation assumption in Airy wave theory, the wave amplitude must be small compared with both the wave length and water depth to make sure non linear effects can be neglected. This can be mathematically expressed as:

$$\frac{H}{\lambda} \ll 1, \qquad \frac{H}{d} \ll 1$$

Where H is the wave height, λ is the wavelength and d is the constant water depth.

6. Irrotationality: Motion of the water particles is assumed to be irrotational which allows the introduction of a velocity potential. This means no vorticity exists in the wave motion because turbulence produced at the seabed can't propagate far into the water column.

The velocity potential can be defined as:

$$u_x = \frac{\partial \Phi}{\partial x}, \qquad u_z = \frac{\partial \Phi}{\partial z}$$

Substituting into continuity gives Laplace's equation:

$$\frac{\partial^2 \Phi}{\partial x^2} + \frac{\partial^2 \Phi}{\partial y^2} + \frac{\partial^2 \Phi}{\partial z^2} = 0$$

Airy wave theory assumes small amplitude, and linearises the governing equations, however non linear wave theories such as Stokes or cnoidal theory account for finite amplitude effects where $\frac{H}{\lambda}$ or $\frac{H}{d}$ are no longer $\ll 1$. Non linear theories capture higher order interactions which are neglected in linear theory however are mathematically complex making linear theory preferable for applications where amplitudes are small.

2.2. Wave Loading on Monopile

To find the time varying wave load on the met mast which is supported by a monopile, the flow is modelled using linear water theory and then the Morison equation is used to express the total horizontal wave force as a function of time, all derivations and calculations are based on formulae available in [15], [16]. The problem is shown in figure 12 which has wave height (H) of 3m, diameter (D) of 7.5m, water depth (d) of 160m and time period (T) of 15 seconds.

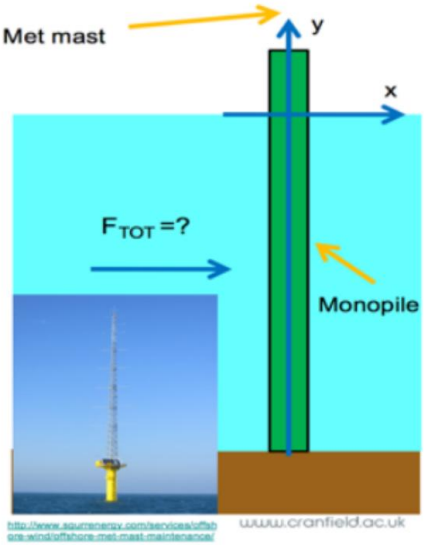


Figure 12 - Wave Loading Problem Definition

2.2.1. Morison Equation Validity, Wave Type and Loading Regime

Firstly to confirm the validity of the Morison equation in the given scenario, $\frac{D}{\lambda} > 0.2$ must be satisfied, where D is the monopiles diameter and wavelength (λ) is defined as $\lambda = \frac{T^2 g}{2\pi}$.

$$\lambda = \frac{T^2 g}{2\pi} = \frac{15^2 * 9.81}{2\pi} = 351.3m$$

$$\frac{D}{\lambda} = \frac{7.5}{351.3} = 0.021$$

$\frac{D}{\lambda}$ is less than 0.2 and therefore the Morison equation is valid to use. Next the wave type must be determined as the kinematics depend on the water depth, this can be done using the following relations:

$$Deep\ water \rightarrow \frac{d}{\lambda} \geq 0.5, \qquad Intermediate \rightarrow 0.05 < \frac{d}{\lambda} < 0.5, \qquad Shallow \rightarrow \frac{d}{\lambda} \leq 0.05$$

For this scenario $\frac{d}{\lambda} = 0.46$, this is less than 0.5m hence strictly this is a intermediate depth condition, however the question explicitly requests deep water linear theory so this will be used. A estimated error is calculated from the ratio of velocity at the free surface for both deep and intermediate waves, where k is the wave number defined as $\frac{2\pi}{\lambda}$:

$$\frac{u_{int}(0,t)}{u_{deep}(0,t)} = \frac{a\omega coth(kd)cos(kx - \omega t)}{a\omega cos(kx - \omega t)} = \coth(kd) = \coth(2.86) = 1.0065$$

This shows an approximate error of 0.65% which is acceptable. Next to determine an appropriate drag coefficient (C_d) and inertia coefficient (C_m) the Keulegan-Carpenter number (K_c) is calculated to determine the wave loading regime and can be expressed as:

$$K_c = \frac{U_m T}{D}$$

Where U_m is the max normal water particle velocity, this can be derived from the velocity potential. The velocity potential for finite depth water waves can be expressed as:

$$\Phi = ac\frac{\cosh\left[k(z+d)\right]}{\sinh\left(kd\right)}\cos\left(kx - \omega t\right)$$

Where a is the wave amplitude expressed as $H/2$, and c is the wave speed expressed as $\frac{\lambda}{T}$. The $\frac{\cosh\left[k(z+d)\right]}{\sinh\left(kd\right)}$ term accounts for the vertical structure with a bottom at $z = -d$ and free surface at $z = 0$. For deep water as d increases the influence of the seabed on wave movement and velocity vanishes so $d \rightarrow \infty$ is assumed, giving $\sinh(kd) \approx \frac{1}{2}e^{kd},\ and\ \cosh[k(z+d)] \approx \frac{1}{2}e^{k(z+d)}$ which results in:

$$\frac{\cosh\left[k(z+d)\right]}{\sinh\left(kd\right)} \approx \frac{\frac{1}{2}e^{k(z+d)}}{\frac{1}{2}e^{kd}} \approx e^{kz}$$

This therefore results in the deep water velocity potential simplifying in a way which is independent of any bottom boundary:

$$\Phi = -ace^{kz}\cos(kx - \omega t)$$

From the definition of velocity potential seen in section 2.1 the horizontal velocity can be found by taking the partial derivative of the velocity potential with respect to x:

$$u_x = \frac{\partial}{\partial x}[-ace^{kz}\cos(kx - \omega t)] = acke^{kz}\sin(kx - \omega t)$$

Substituting:

$$ack = \left(\frac{H}{2}\right)\left(\frac{gT}{2\pi}\right)\left(\frac{4\pi^2}{gT^2}\right) = \frac{H\pi}{T}, \qquad \rightarrow u_x = \frac{H\pi}{T}e^{kz}\sin(kx - \omega t)$$

From this U_m can be determined as u_x is max when $\sin(kx - \omega t) = 1$ and $z = 0$ as z is positive upwards, this results in a U_m of 0.628m/s, when substituting into the K_c equation a K_c of 1.26 is obtained. Combined with the condition of $\mathrm{H/D} < 1.5$ which is also satisfied, the wave loading regime can be seen as inertia dominated. This gives a C_m of 2 and a C_d of 0, this essentially makes the drag component of the Morison equation negligible.

2.2.2. Horizontal Wave Force Derivation and Loading Calculations

The Morison equation can be expressed as:

$$F = 0.5\varrho C_d A_p V|V| + C_m \varrho V_{vol}\frac{\partial u}{\partial t}$$

The $0.5\varrho C_d A_p V|V|$ term represents the drag forces and the $C_m \varrho V_{vol}\frac{\partial u}{\partial t}$ term the inertia forces. Since C_m is zero the drag term cancels out leaving the total force equal to the inertia force. Horizontal acceleration can then be calculated by taking the partial derivative of u_x with respect to time:

$$a_x = \frac{\partial u}{\partial t} = \frac{H\pi}{T}e^{kz} * \frac{\partial}{\partial t}[\sin(kx - \omega t)] = -\frac{2H\pi^2}{T^2}e^{kz}\cos(kx - \omega t)$$

The maximum wave induced horizontal inertia force on the structural member can be expressed as:

$$F_I = \varrho C_M \frac{\pi D^2}{4}\int_{-d}^{0} \dot{u} dz$$

Substituting a_x gives the equation for total horizontal wave force as a function of time:

$$F_I = -\varrho C_M \frac{\pi D^2}{4}\frac{2H\pi^2}{T^2}\cos(kx - \omega t)\int_{-d}^{0} e^{kz} dz$$

$$\therefore \int_{-d}^{0} e^{kz} dz = \frac{1}{k}e^{kz}|_{-d}^{0} = \frac{1}{k}e^{k*0} - \frac{1}{k}e^{-kd} = \frac{1}{k}(1 - e^{-kd})$$

$$\therefore F_I = -\varrho C_M \frac{\pi D^2}{4}\frac{2H\pi^2}{T^2}\frac{1}{k}(1 - e^{-kd})\cos(kx - \omega t)$$

Substituting in known values gives:

$$F_I = -(1025)(2)\left(\frac{\pi * 7.5^2}{4}\right)\left(\frac{2 * 3 * \pi^2}{15^2}\right)\left(\frac{1}{\frac{2\pi}{351.29}}\right)\left(1 - e^{-\left(\frac{2\pi}{351.29} * 160\right)}\right)\cos\left(kx - \omega t\right)$$

$$\therefore F_I = -1.256\cos(kx - \omega t)\, MN$$

Maximum horizontal inertia force is when $\cos(kx - \omega t) = -1$, so $F_{Imax} = 1.256\, MN$

Therefore, the total maximum load on the met mast supported monopole is $1.256\, MN$.

3. References

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